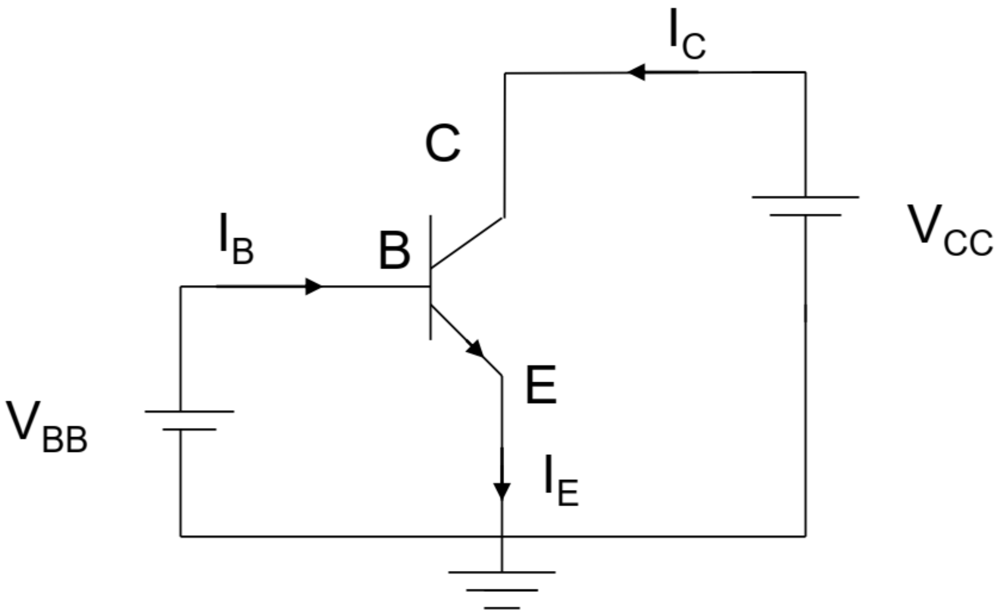
CE Transistor - npn



Common Emitter (CE) Transistor

β = Common emitter current amplification factor

$= I_C/I_B$

Current Gain of the Transistor

$$I_C = \beta I_B + (\beta+1) I_{CO}$$

$$\beta = \alpha/(1-\alpha)$$

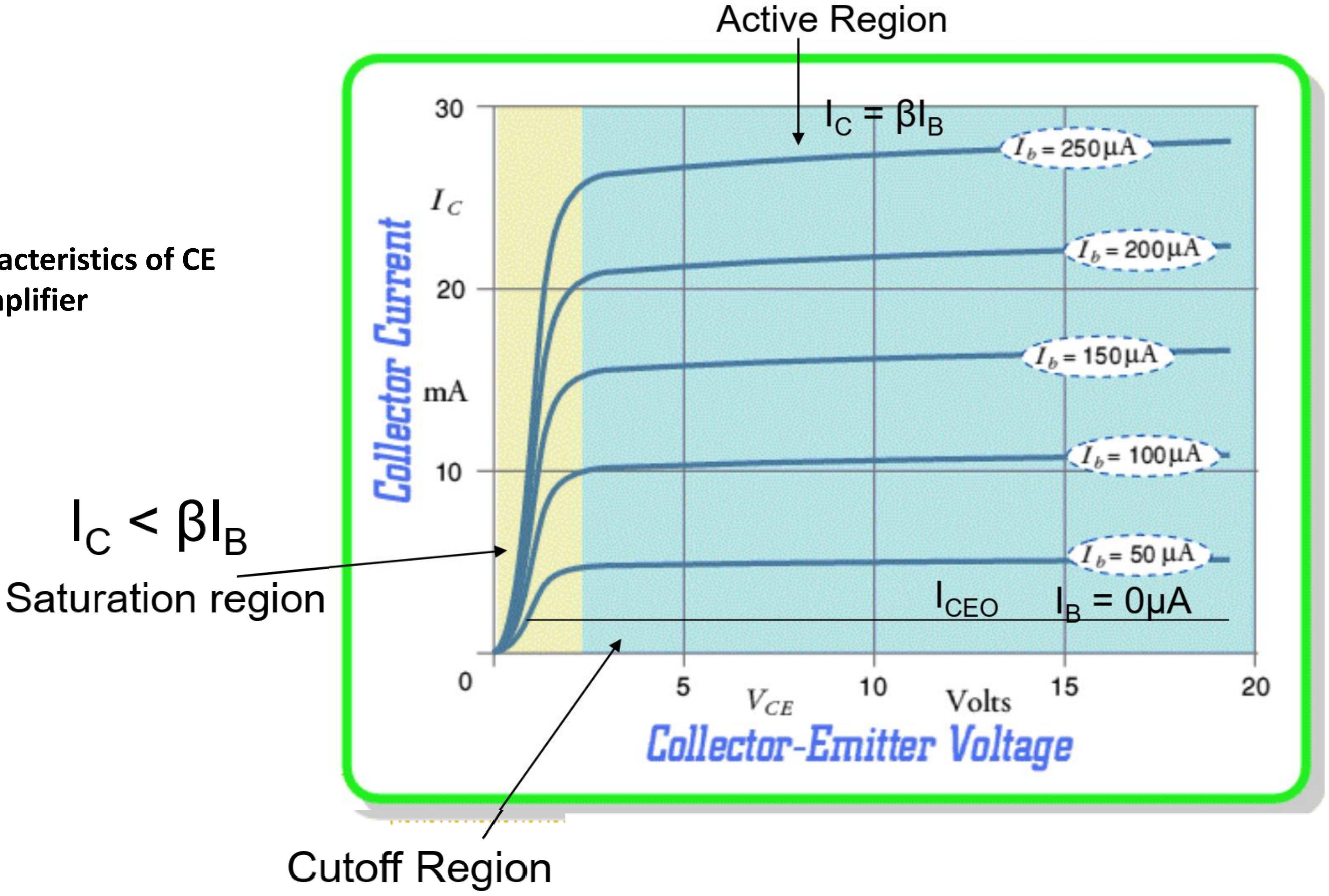
Example -

For $\alpha = 0.98$,

$\beta = 49$

β increases rapidly as $\alpha \rightarrow 1$

Output characteristics of CE Amplifier



Transistor in Cutoff Region

Both the BC and the BE junctions are reverse biased

$I_E=0$ $I_C=I_{CO}\approx 0$ $I_B=-I_C=-I_{CO}\approx 0$

Transistor in Saturation Region

Both the BC and BE junctions are forward biased and $I_C<\beta I_B$

$V_{BE}\approx 0.7\ V$ $V_{BC}\approx 0.6\ V$ $V_{CE}\approx 0.1$

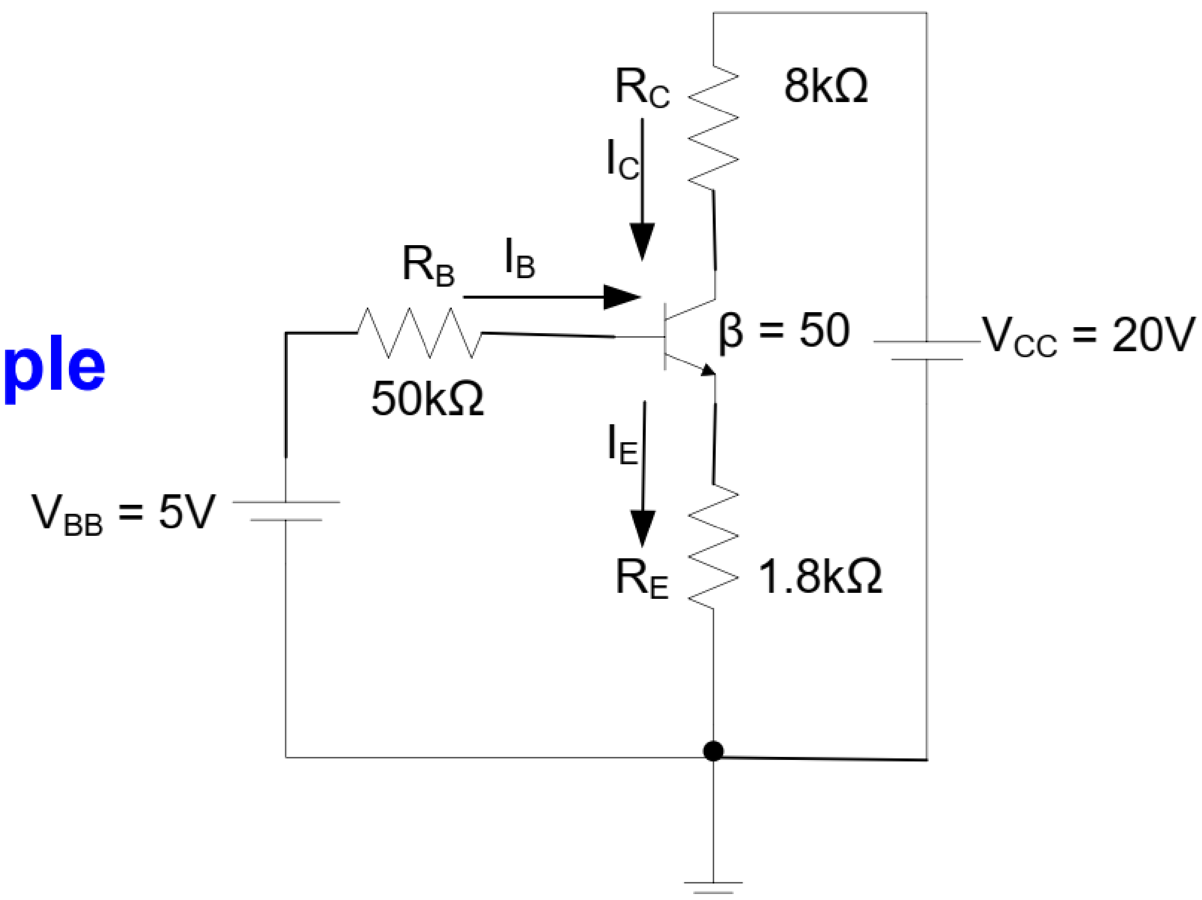
Transistor in Active Region

The BE junction is forward biased but the BC junction is reverse biased and $I_C \approx \beta I_B$

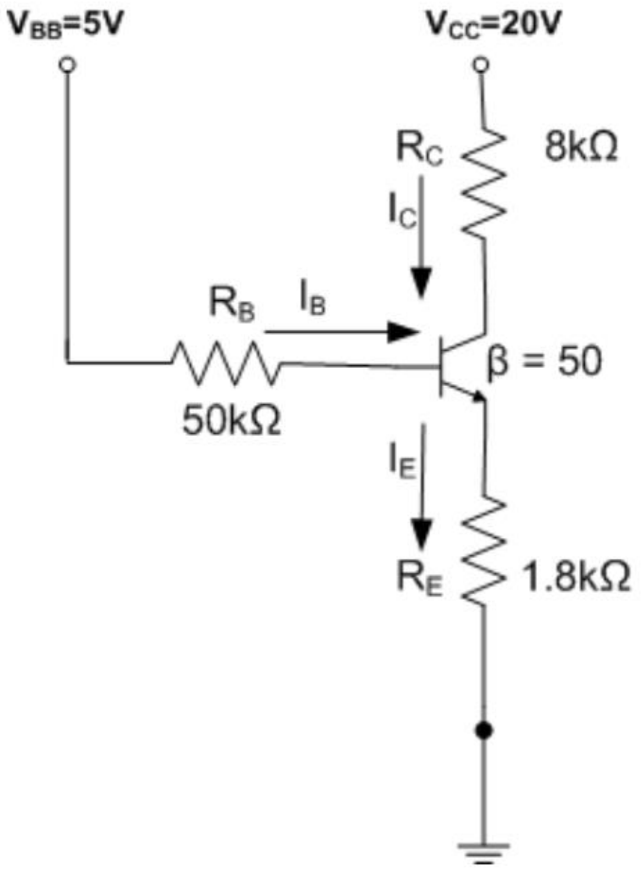
$V_{BE}=0.7$ V

Transistor is operated in this region when we want to use it as an amplifier!

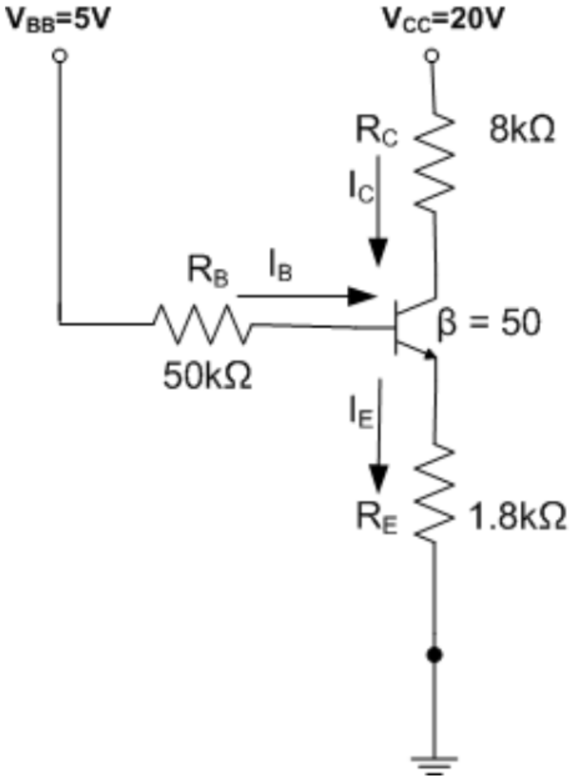
Example



R_C
$8k\Omega$
I_C
R_B
I_B
$\beta = 50$
$V_{CC} = 20V$
$50k\Omega$
I_E
$V_{BB} = 5V$
R_E
$1.8k\Omega$



V_{BB}=5V
V_{CC}=20V
R_C
$8k\Omega$
I_C
R_B
I_B
$\beta = 50$
$50k\Omega$
I_E
R_E
$1.8k\Omega$



$$I_c = \beta I_B = 50 I_B \qquad I_E = I_c + I_B = 51 I_B$$

$$\boxed{\begin{array}{l} I_B = \dfrac{V_{BB} - V_{BE}}{R_B + (\beta + 1) R_E} = \dfrac{5 - 0.7}{50 + 51 * 1.8} = 0.0303\ mA \\ I_C = 50 * 0.0303 = 1.52\ mA \qquad I_E = 1.55\ mA \end{array}}$$

$$V_E = I_E R_E = 1.55 * 1.8 = 2.79\ V$$

$$V_B = V_E + V_{BE} = 2.79 + 0.7 = 3.49\ V$$

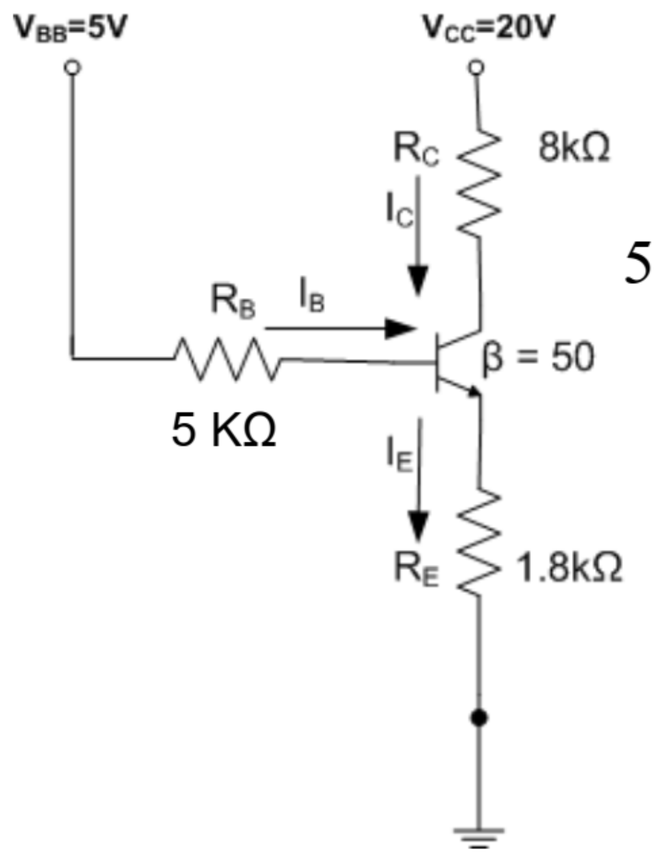
$$V_C = V_{CC} - I_C R_C = 20 - 1.52 * 8 = 7.84$$

$$\boxed{V_{CE} = V_C - V_E = 7.84 - 2.79 = 5.05\ V}$$

Note that V_{BC} = 3.49 -7.84 = -4.35 V, i.e. the B-C junction is reverse biased, implying that the transistor is working in the Active Mode

The Bias Point is specified by (V_{CE}, I_C, I_B)

What would happen if we reduced the base resistance to 5KΩ?



Let's assume that the transistor is still in the active region so $I_C=\beta I_B$ and $I_E=(\beta+1)I_B$. Then -

$$5-0.7=5I_B+51I_B(1.8) \quad \Rightarrow \quad I_B=\frac{4.3}{5+51\times1.8}=0.0444 \text{ mA}$$

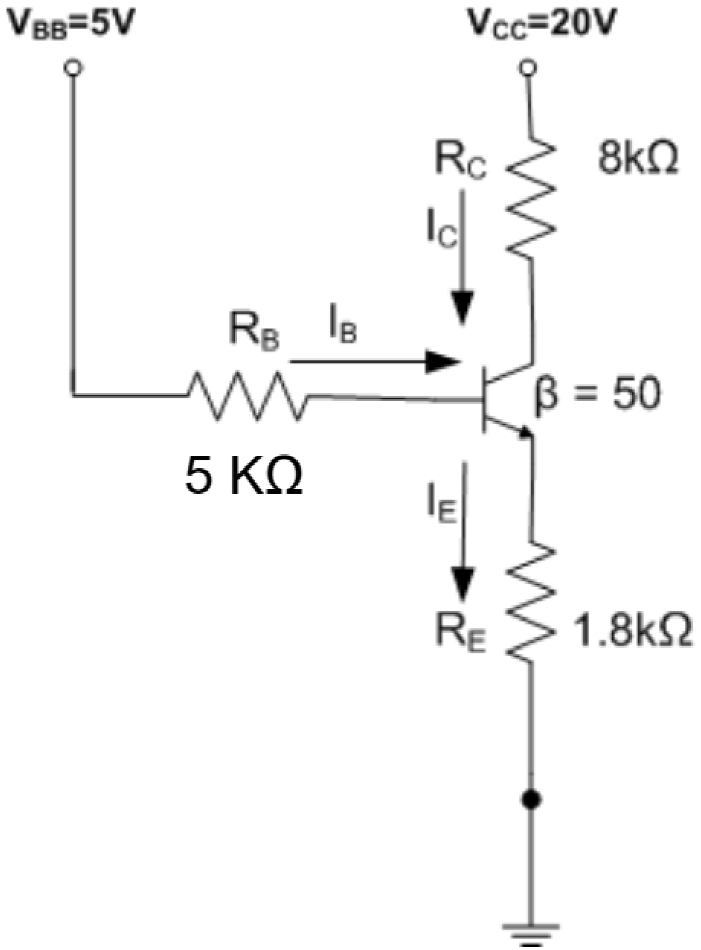
$I_C=50I_B$ = 2.22 mA $\qquad$ $I_E=51I_B$ = 2.2644 mA

$V_E=I_ER_E$ = 4.076V

$V_B=V_E+0.7$= 4.776V

$V_C=V_{CC}-I_CR_C$ = 20 – $8I_C$ = 2.24V

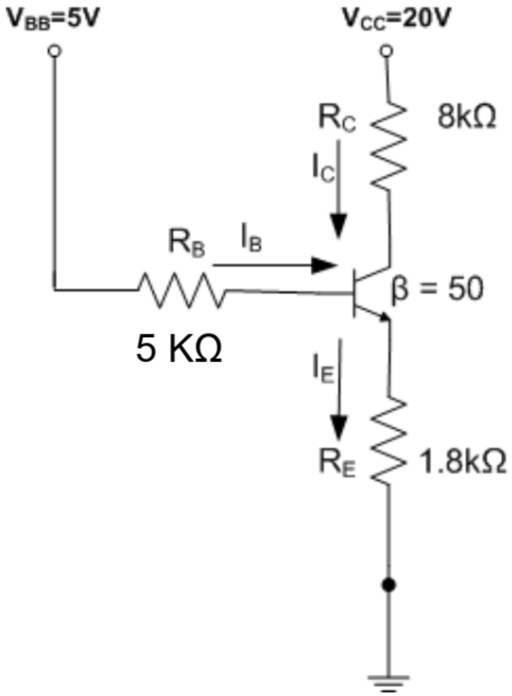
Note that B-C junction is then forward-biased which violates our earlier assumption that the transistor is in the active region. Therefore, our earlier assumption is WRONG – the transistor will not be in the active region



Can the transistor be in CUTOFF?

If the transistor is in CUTOFF then $I_B=I_C=I_E=0$. Therefore $V_B=5V$ and $V_E=0V$ but that will forward bias B-E which violates the condition for transistor to be in CUTOFF. Therefore, the transistor cannot be in CUTOFF.

What would happen if we reduced the base resistance to 5KΩ?



Can the transistor be in Saturation?

In saturation, $V_{CE(sat)}$ =0.1V and V_{BE}=0.7V

Note that $I_C=\beta I_B$ will not hold in saturation, but we will still have $I_E=I_B+I_C$

$$+5 = 5I_B + 0.7 + 1.8(I_B + I_C) \quad \Rightarrow \quad 6.8I_B + 1.8I_C = 4.3$$

$$+20 = 8I_C + 0.1 + 1.8(I_B + I_C) \quad \Rightarrow \quad 1.8I_B + 9I_C = 19.1$$

Solving - I_B=0.0745 mA, I_C=2.107 mA & I_E=2.182 mA

Note that

(a) $I_C<\beta I_B$

(b) V_E=3.93 V V_B=4.63 V V_C=4.03 V

So B-C is forward biased

The transistor is indeed in saturation

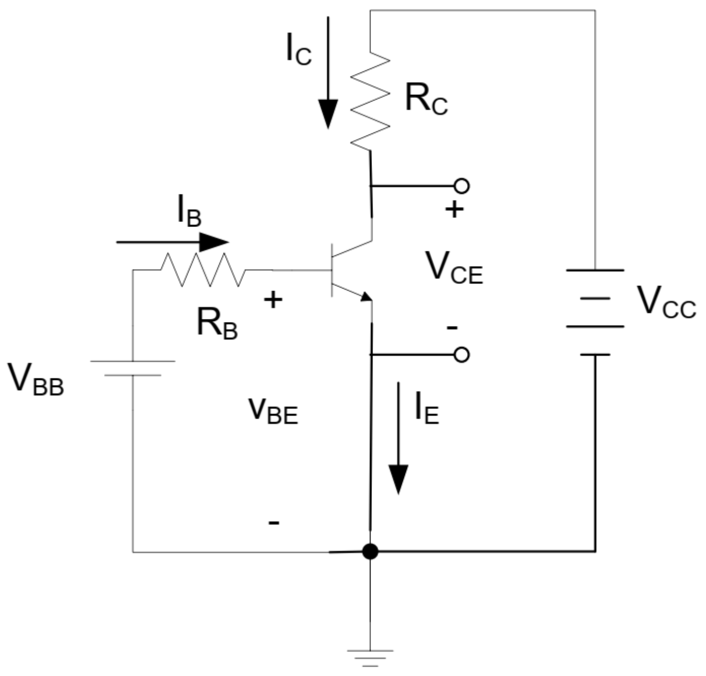
Biasing the Transistor

To operate as an amplifier a transistor must be biased in the active region. In that case, the transistor can amplify a *small ac signal* faithfully.

Bias Point (or Quiescent Operating Point, i.e. Q-Point) is decided by the DC voltages applied and the values of the resistances used. (See earlier example.)

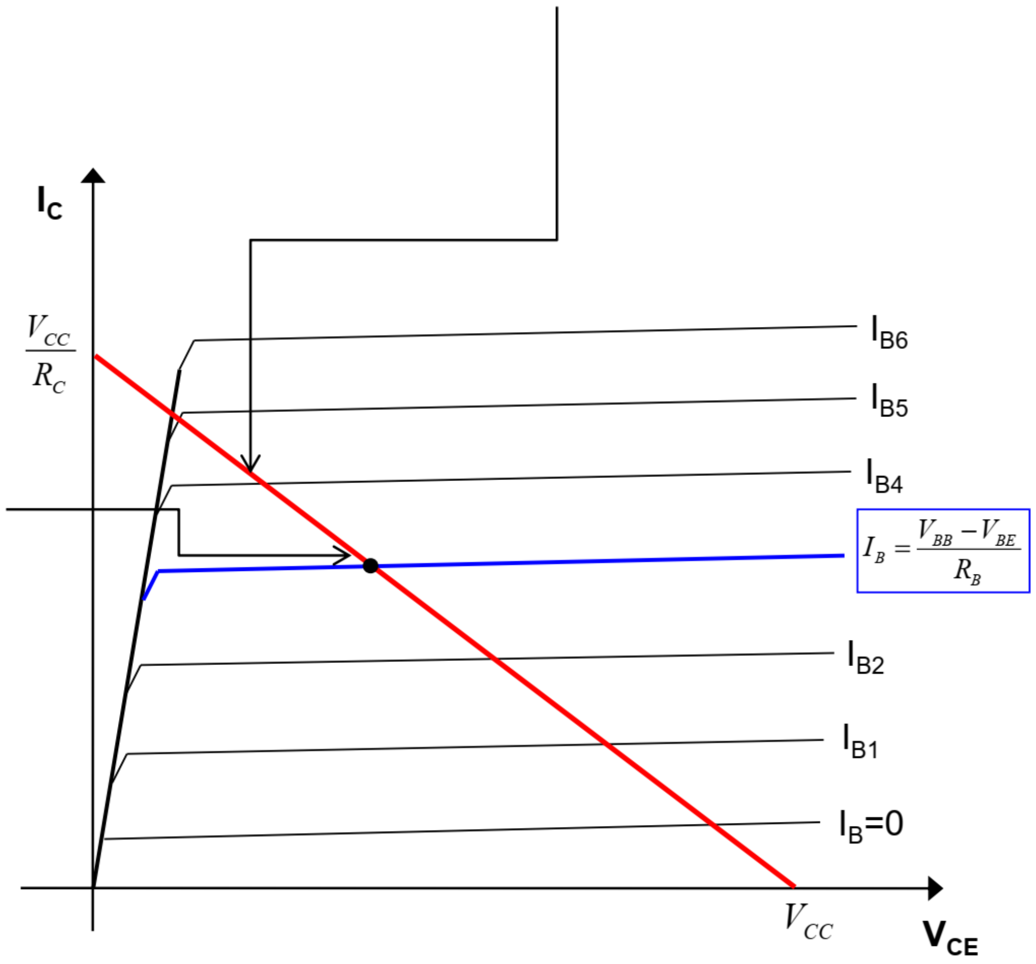
The Bias Point is specified by the three quantities $\mathbf{V_{CE}}$, $\mathbf{I_C}$ **and** $\mathbf{I_B}$. *This point must be in the active region of the transistor if we want to use the transistor (with that bias point) as an amplifier.*

Biasing the Transistor



$$I_C = \frac{V_{CC} - V_{CE}}{R_C} \qquad I_C = -\left(\frac{1}{R_C}\right)V_{CE} + \frac{V_{CC}}{R_C}$$

Plot of I_C vs. V_{CE} is called the 'Load Line"



Intersection of the load line with the I_C-V_{CE} characteristic for the given value of I_B decides the bias point.